

Operator Discovery Report: select_transfer_scaffold_by_structure

- Operator type: scaffold_selector.
- Host scaffold: nearest_neighbor_2opt.
- Core idea: Deterministically picks a TSP baseline scaffold based on instance structure descriptors (bottleneck/cluster/grid/elongation) to improve transfer while keeping local 2-opt style behavior consistent.
- Novelty classification: descriptor_based_scaffold_selector.
- Rediscovery signature: scaffold_selector:random_restart_2opt:clustered_local_search:clustered_local_search.
- Surviving candidate: False.
- Problem class where it helps: heldout_tsplib, two_cluster_bottleneck_tsp.

Pseudocode

- Read instance descriptors to classify the layout family.
- Dispatch the instance to the preferred scaffold for that structure class.
- Fall back to a safe default scaffold when the structure is ambiguous.

Complexity

- Non-empty lines: 13.
- Complexity score: 0.46.
- Fingerprint: e04c01207bedeb3a.

Why It Should Help

- The operator selects among simple solvers from instance descriptors, so any gain should come from a reusable structure-to-scaffold mapping.

Failure Cases

- Underperforms on nearest_neighbor_trap_tsp with gap delta 0.034212.
- Underperforms on uniform_euclidean with gap delta 0.013605.
- Does not transplant cleanly into cheapest_insertion_2opt.
- Runtime inflation is high at 0.981692.

Transplant Results

- nearest_neighbor_2opt gap delta -0.059029, runtime delta 291.734943, distance-eval delta 110250.142857.
- cheapest_insertion_2opt gap delta 0.048293, runtime delta -4282.379157, distance-eval delta -6650807.857143.
- random_restart_2opt gap delta -0.038703, runtime delta 40.549529, distance-eval delta 94468.571429.
- sparse_three_opt gap delta -0.057743, runtime delta 315.845714, distance-eval delta 106828.857143.
- clustered_local_search gap delta -0.086334, runtime delta 114.2281, distance-eval delta 95342.428571.

Instance-Family Results

- heldout_tsplib operator gap 0.182167 vs baseline 0.235058.
- clustered_tsp operator gap 0.113212 vs baseline 0.110853.
- elongated_corridor_tsp operator gap 0.006647 vs baseline 0.003204.
- grid_like_tsp operator gap 0.106696 vs baseline 0.106696.
- nearest_neighbor_trap_tsp operator gap 0.034212 vs baseline 0.0.
- two_cluster_bottleneck_tsp operator gap 0.001685 vs baseline 0.109057.
- uniform_euclidean operator gap 0.013605 vs baseline 0.0.

Pareto Results

- Gap delta: -0.02318.
- Runtime inflation: 0.981692.
- Distance-evaluation inflation: 11.735255.
- Same-gap-faster flag: False.
- Better-gap-same-runtime flag: False.

Ablation Results

- disabled mean gap 0.121317 and delta vs full operator 0.021933.
- flattened_selector mean gap 0.121317 and delta vs full operator 0.021933.
- shuffled_selector mean gap 0.081893 and delta vs full operator -0.017491.